

The empirical recurrence for OEIS A253330: recovering a conjecture that is not written down

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Abstract

OEIS A253330 records “Empirical recurrence of order 63”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 676$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 63, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 63 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A253330 is “Number of $(5+1)X(n+1)$ 0..1 arrays with every $2X2$ subblock sum nondecreasing horizontally, vertically and antidiagonally ne-to-sw.”. It has offset 1 and begins

676, 2042, 6190, 19312, 58505, 173903, 503944, 1397185, ...

The entry states:

Empirical recurrence of order 63 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:58:16, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 676$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 676$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 1352$ exact terms of the model returns order 63 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{63} c_i a(n-i),$$

c_1	4	c_{17}	219635712	c_{33}	30399541644	c_{49}	−514808256
c_2	36	c_{18}	−111911552	c_{34}	−51286804500	c_{50}	−1619296704
c_3	−162	c_{19}	−689846188	c_{35}	−28786717410	c_{51}	277582464
c_4	−604	c_{20}	456557112	c_{36}	55865954980	c_{52}	489565056
c_5	3154	c_{21}	1846825596	c_{37}	23043376130	c_{53}	−107579136
c_6	6186	c_{22}	−1475810036	c_{38}	−52927725990	c_{54}	−118077184
c_7	−39306	c_{23}	−4244002060	c_{39}	−15237886650	c_{55}	31074304
c_8	−42019	c_{24}	3939977274	c_{40}	43528988089	c_{56}	21847296
c_9	352234	c_{25}	8411721276	c_{41}	7928932874	c_{57}	−6606336
c_{10}	185034	c_{26}	−8879451716	c_{42}	−30968624214	c_{58}	−2911744
c_{11}	−2417268	c_{27}	−14420754064	c_{43}	−2828143692	c_{59}	984064
c_{12}	−373572	c_{28}	17114577864	c_{44}	18960199036	c_{60}	248832
c_{13}	13210560	c_{29}	21404362296	c_{45}	237170744	c_{61}	−92160
c_{14}	−1570272	c_{30}	−28443491112	c_{46}	−9916598664	c_{62}	−10240
c_{15}	−59024568	c_{31}	−27477002760	c_{47}	560142864	c_{63}	4096
c_{16}	19656918	c_{32}	40968113595	c_{48}	4386822816		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 63, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $63 + 4$ terms removed returns order 63 as well, so $\deg q_{\min} = 63 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $63 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 63 that this sequence satisfies — and that family turns out to have exactly one member.

References

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